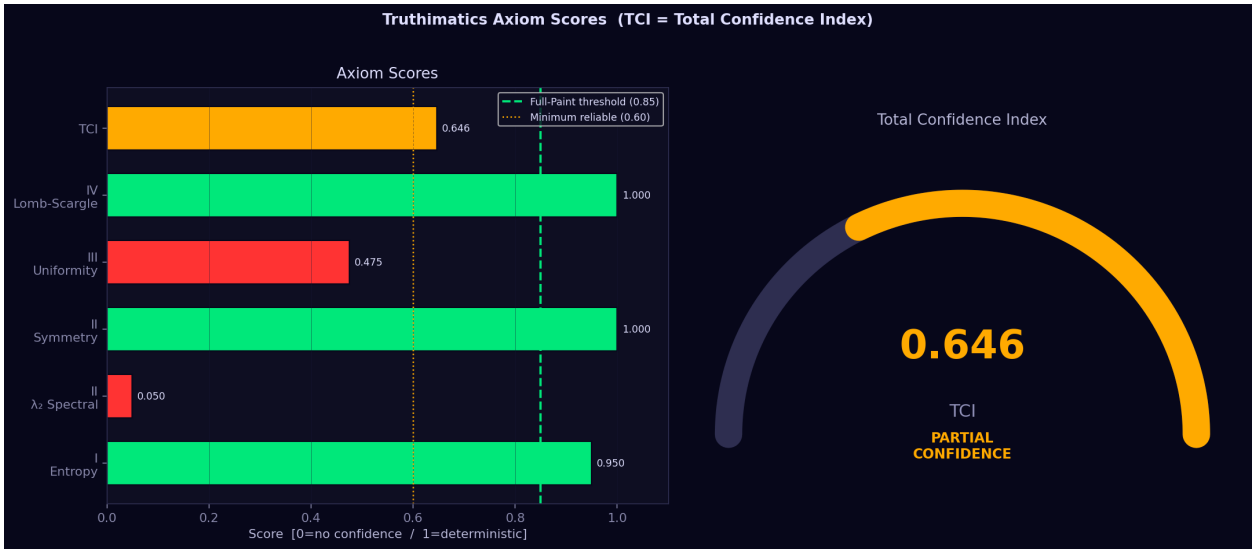
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_3AA9257757
Taxa	Protoceratops
Clade	Ornithischian
PBDB Records	38
Mass (regression)	3819.1 kg (Campione & Evans 2012)
Mass (volumetric)	4065.3 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	4.279 m³
Validity Score	1.0000 / 1.0000
TCI	0.6462 (partial)
Axiom I Entropy	0.9495
Axiom II λ■	0.1992 (symmetry 100%)
Axiom III Uniformity	0.4748
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	68.0% Measured / 32.0% Inferred
Elements	25
CoG (m)	[0.287, 0.982, 0.0]
Generated	2026-05-01 00:02 UTC

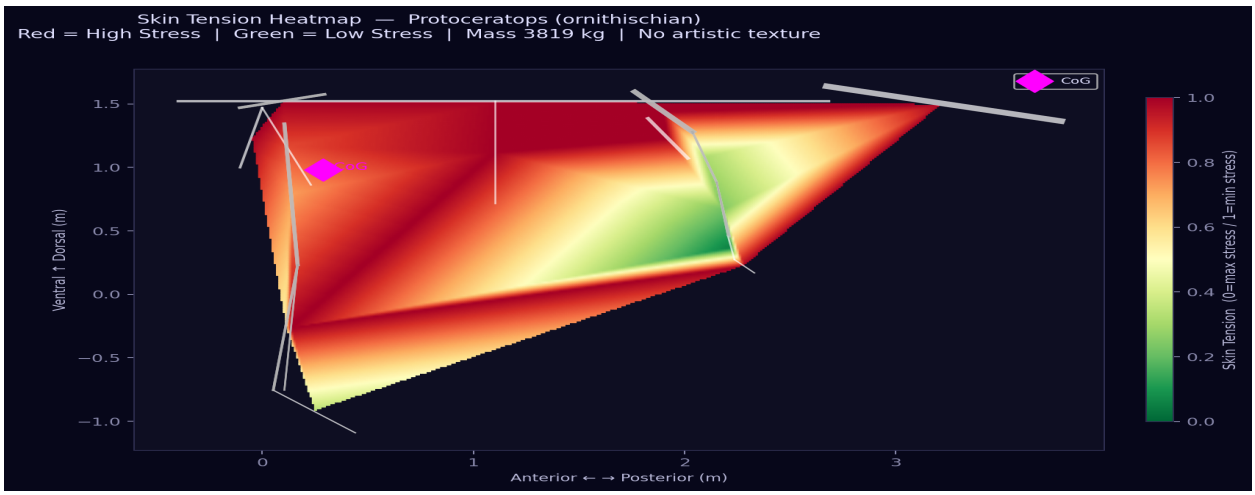
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



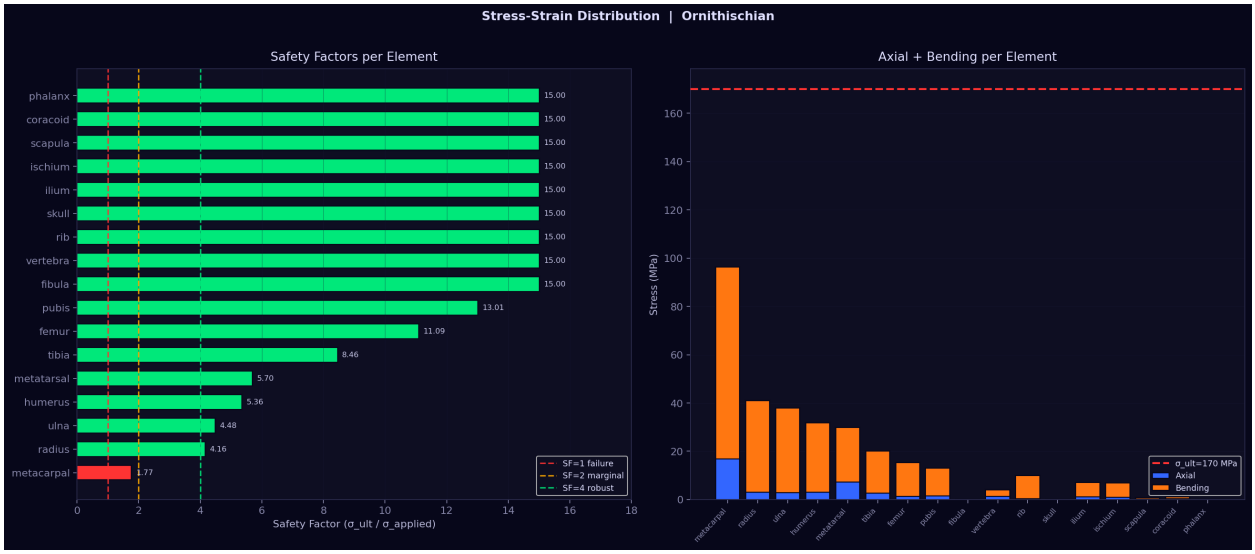
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

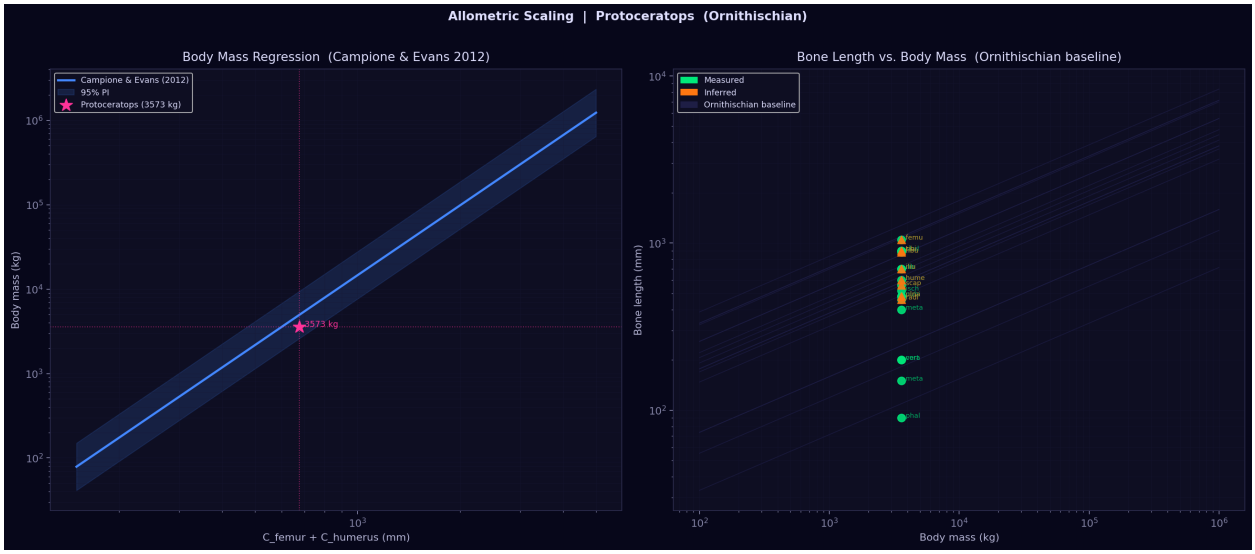


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.391	0.749	1.140	15.000	OK
femur	1.294	14.034	15.328	11.091	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	3.028	28.689	31.716	5.360	OK
ilium	1.116	5.993	7.109	15.000	OK
ischium	0.945	5.865	6.810	15.000	OK
metacarpal	16.777	79.450	96.227	1.767	CRITICAL
metatarsal	7.232	22.591	29.823	5.700	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	1.563	11.506	13.068	13.008	OK
radius	3.063	37.821	40.883	4.158	OK
rib	0.298	9.588	9.886	15.000	OK
scapula	0.158	0.541	0.699	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	2.701	17.401	20.102	8.457	OK
ulna	2.832	35.084	37.916	4.484	OK
vertebra	1.294	2.673	3.968	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.35 m	Femur + tibia + metatarsal
Estimated Speed	14.6 km/h (4.06 m/s)	Gait: Trot | Fr=0.715
Stride Length	4.875 m	Alexander (1976) / Thulborn (1990)
Bite Force	17493.1 N (1783.2 kgf)	Ornithischian regression (Norman & Weishampel 1991)
Skull Length	900.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.389	Circumference / length (dimensionless)
Limb Strength Index	4e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	9.43 m	Allometric from femur length
Est. Neck Length	1.04 m	Allometric from femur length
Est. Tail Length	2.64 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	200.0	140.0	1570.17	1960.0	Measured	0.88	15.00
femur	1050.0	130.0	7107.81	8872.7	Measured	0.88	11.09
femur_right	1050.0	130.0	7107.81	8872.7	Symmetry	0.84	11.09
fibula	880.0	40.0	398.10	417.5	Measured	0.88	15.00
fibula_right	880.0	40.0	398.10	417.5	Symmetry	0.84	15.00
humerus	600.0	85.0	1736.40	2167.5	Measured	0.88	5.36
humerus_right	600.0	85.0	1736.40	2167.5	Symmetry	0.84	5.36
ilium	700.0	140.0	5495.59	6860.1	Measured	0.88	15.00
ilium_right	700.0	140.0	5495.59	6860.1	Symmetry	0.84	15.00
ischium	520.0	90.0	1687.13	2106.0	Measured	0.88	15.00
metacarpal	150.0	35.0	27.42	22.6	Measured	0.88	1.77
metatarsal	400.0	55.0	484.67	605.0	Measured	0.88	5.70
phalanx	90.0	28.0	8.51	6.6	Measured	0.88	15.00
pubis	480.0	70.0	942.10	1176.0	Measured	0.88	13.01
radius	460.0	50.0	460.64	575.0	Measured	0.88	4.16
radius_right	460.0	50.0	460.64	575.0	Symmetry	0.84	4.16
rib	700.0	35.0	343.47	428.8	Measured	0.88	15.00
scapula	560.0	220.0	10856.59	13552.3	Measured	0.88	15.00
scapula_right	560.0	220.0	10856.59	13552.3	Symmetry	0.84	15.00
skull	900.0	400.0	57679.64	72001.5	Measured	0.88	15.00
tibia	900.0	90.0	2920.03	3645.1	Measured	0.88	8.46
tibia_right	900.0	90.0	2920.03	3645.1	Symmetry	0.84	8.46
ulna	480.0	52.0	519.89	649.0	Measured	0.88	4.48
ulna_right	480.0	52.0	519.89	649.0	Symmetry	0.84	4.48
vertebra	200.0	130.0	1353.87	1690.0	Measured	0.88	15.00